

Background-Free Intensity Autocorrelation for Femtosecond X-Ray Pulses

Taito Osaka^{1,*}, Shotaro Matsumura^{2,†}, Masafumi Miyake², Yasuhisa Sano², Ichiro Inoue¹, Yuichi Inubushi^{1,3},
Kensuke Tono^{1,3}, Kenji Tamasaku¹, and Makina Yabashi^{1,3}

¹*RIKEN SPring-8 Center, 1-1-1 Kouto, Sayo-cho, Sayo-gun, Hyogo 679-5148, Japan*

²*Division of Precision Engineering and Applied Physics, Graduate School of Engineering,
The University of Osaka, 2-1 Yamada-oka, Suita, Osaka 565-0871, Japan*

³*Japan Synchrotron Radiation Research Institute, 1-1-1 Kouto, Sayo-cho, Sayo-gun, Hyogo 679-5198, Japan*



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A background-free intensity autocorrelation measurement for 10-keV femtosecond x-ray pulses is demonstrated using x-ray second harmonic generation. The narrow phase-matching condition of x-ray second harmonic generation allows for the selective generation of either the autocorrelation or background signals in a quasicollinear geometry with a crossing angle of ~ 300 μ rad, achieving a geometric time resolution of ~ 1 attosecond. The measured autocorrelation traces exhibit zero background and clearly reveal temporal modifications through an Si(311) monochromator, verifying x-ray temporal shaping with perfect crystals. The applicability of this method to attosecond x-ray pulses is also discussed.

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Ultrafast lasers, with durations ranging from femtoseconds to attoseconds, play crucial roles in modern science and technologies, enabling measurements of ultrafast dynamics in matter [1–3], advanced materials processing [4], and production of exotic states that exist in the Universe [5]. The advent of x-ray free-electron lasers (XFELs) [6–10] has extended the wavelength range down to ~ 1 Å, offering unique capabilities along with the atomic-scale spatial resolution, photon energies (~ 10 keV) comparable to core-level binding energies, and high penetration to optically opaque materials [11]. Recent developments have further enabled attosecond XFEL pulses with peak powers far exceeding other attosecond sources [12,13], opening up a new era in ultrafast science.

Accurate knowledge of the pulse duration of ultrafast lasers is vital for their applications. At optical wavelengths, intensity autocorrelation (IAC) has been widely used to characterize averaged pulse durations [14]. In IAC measurements, a sample pulse is split into two replicas, which are then superimposed on a nonlinear medium—typically through a focusing device—with a variable delay. Finally, the nonlinear signal intensity is measured as a function of the delay. The trace is equivalent to the temporal autocorrelation of the sample pulse, providing a direct estimate of the pulse duration under a presumable assumption of the pulse shape.

At angstrom wavelengths, the situation is entirely different due to significantly lower nonlinear efficiencies and the

lack of optical autocorrelators. Instead, free-electron laser simulations combined with electron-bunch diagnostics [15,16] are employed as a standard diagnostic method for temporal profiles of XFEL pulses. Although this method is reliable when the XFEL is operating far from the single-spike limit, this approach struggles to capture the *final* pulse shape after beamline instruments, especially monochromators. Complementally, Fourier transformation of XFEL spectra [17,18] and intensity-correlation measurements of x-ray fluorescence [19] have been reported. These optical methods can provide the final temporal properties, requiring nontrivial assumptions. Optical streaking of photoelectrons can retrieve single-shot profiles from a few femtoseconds to attosecond regime without such assumptions [12], yet no reports exist for hard x-ray pulses.

Recently, a two-photon absorption (TPA) IAC measurement was demonstrated for 9-keV XFEL pulses [20]. Although the measured trace reasonably agreed with simulations, background signals generated by individual subpulses and large statistical errors prevented reliable estimation of pulse duration and shape. An optimistic estimate suggests that detecting more than 10 000 background signal photons is required for detailed characterization. Achieving background-free detection would reduce the required accumulations by an order of magnitude, which is crucial for probing the inherently weak nonlinear processes at angstrom wavelengths.

Mainly two approaches have been demonstrated at optical wavelengths to eliminate background signals: (1) operating in a crossing geometry and (2) using pulse replicas with orthogonal polarizations and a birefringent nonlinear medium. Both methods employ a wave mixing process, such as second harmonic generation (SHG), in

*Contact author: osaka@spring8.or.jp

†Present address: S-Surface Technologies Co., Ltd., 2-5-38 Saito-Yamabuki, Ibaraki, Osaka 567-0086, Japan.

which the three associating waves—pulse replicas 1 and 2, and the nonlinear signal—must satisfy the phase-matching condition $\mathbf{k}_1 + \mathbf{k}_2 = \mathbf{k}_{\text{nonlin}}$, where $\mathbf{k}$ denotes the internal wave vector of each wave, as indicated by the subscript. Since both approaches ensure $\mathbf{k}_1 \neq \mathbf{k}_2$, the autocorrelation signal is generated on a different angle from the background signal by the individual replicas, which satisfy $2\mathbf{k}_i = \mathbf{k}_{\text{nonlin}}$ with $i = 1$ or 2 .

In the first approach, the crossing angle is typically on the order of degrees, allowing the spatial separation of nonlinear signals. However, this method degrades the time resolution due to variations in the delay across the superimposed area. As a result, it has been applied primarily to pulses with durations longer than 100 fs. While the second approach allows operation in a collinear geometry, it is impractical at angstrom wavelengths due to small birefringence of materials at such short wavelengths.

In this Letter, a method for eliminating the background signal in IAC measurements is demonstrated while minimizing the impact on time resolution. X-ray SHG was measured through the diamond 220 reflection [21] in a quasicollinear geometry with a geometric time resolution of ~ 1 as. The narrow phase-matching condition of x-ray SHG enables complete background suppression even at a sub-milliradian crossing angle. The measured autocorrelation traces exhibit zero background and show a clear distinction between measurements with and without an additional Si(311) monochromator. For the Si(311) case, the autocorrelation trace reasonably agrees with the impulse response of the beamline optics, verifying x-ray temporal shaping using perfect crystals. The feasibility of extending this method to the characterization of attosecond XFEL pulses is also discussed.

As reported in [21], phase matching of x-ray SHG cannot be achieved in the forward direction at angstrom wavelengths unlike optical regimes. Instead, it can be satisfied by using the periodic nature of the electron density in crystals. Figure 1 shows a conceptual illustration of the phase matching with crossed pulse replicas. Here, we can realize that the phase matching is achieved under the following three conditions using the definition in Fig. 1:

$$\mathbf{k}_\omega^{(1)} + \mathbf{k}_\omega^{(2)} + \mathbf{G} = \mathbf{k}_{2\omega}^{(\text{ac})}, \quad (1a)$$

$$2\mathbf{k}_\omega^{(1)} + \mathbf{G} = \mathbf{k}_{2\omega}^{(\text{bg1})}, \quad (1b)$$

$$2\mathbf{k}_\omega^{(2)} + \mathbf{G} = \mathbf{k}_{2\omega}^{(\text{bg2})}, \quad (1c)$$

where $\mathbf{G}$ is the reciprocal lattice vector. Under the first condition, both pulse replicas contribute to the SHG process, which depends on the time delay between the two pulses. On the other hand, since only one of the two pulses contributes under the other conditions, they are independent on the time delay and can form a constant background on measured autocorrelation traces. However,

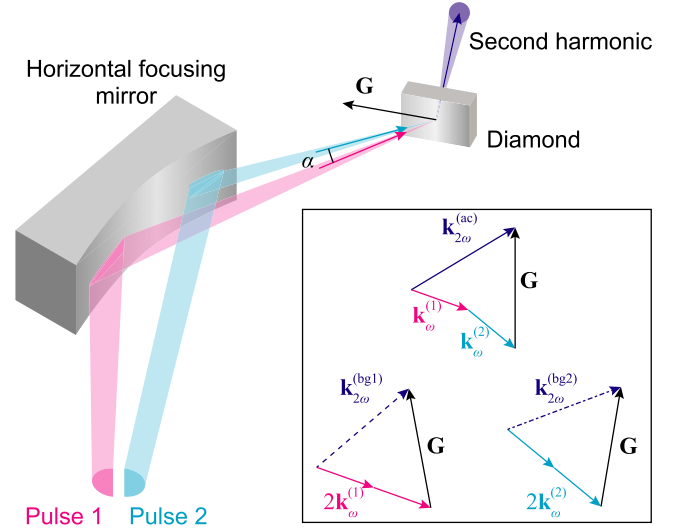


FIG. 1. Schematics of the second-harmonic generation setup with crossing pulse replicas with a crossing angle α (not scaled). $\mathbf{G}$ is the reciprocal lattice vector of the diamond (220) plane. Inset shows schematic diagrams of three phase-matching conditions. $\mathbf{k}_\omega^{(1)}$, $\mathbf{k}_\omega^{(2)}$, and $\mathbf{k}_{2\omega}$ denote the internal wave vectors of Pulse 1 and Pulse 2, and the second-harmonic wave, respectively. Top diagram represents the autocorrelation signal [$\mathbf{k}_{2\omega}^{(\text{ac})}$] and bottom two constant background signals generated by either Pulse 1 [$\mathbf{k}_{2\omega}^{(\text{bg1})}$] or Pulse 2 [$\mathbf{k}_{2\omega}^{(\text{bg2})}$].

if the width of the rocking curve, that is, the SHG efficiency as a function of the crystal angle, is narrower than the angular separation between each condition, we can selectively generate only the autocorrelation signal through Eq. (1a) by adjusting the crystal angle.

The angular separation can be expressed as $\alpha/2$, where α is the crossing angle. A large α facilitates the exclusion of the background whereas the time resolution is geometrically degraded, as mentioned above. A geometric time resolution can be described by $(w \tan \alpha)/c$, where w is the beam size along the crossing direction and c is the speed of light. For example, the geometric time resolution becomes ~ 60 fs with $w = 1$ mm and $\alpha = 1^\circ$. At angstrom wavelengths, both a narrow rocking curve width of $100\text{--}150$ μrad and a small beam size of ~ 1 μm can be achieved [21,22]. The narrow width allows for the use of a small α of ~ 300 μrad , resulting in a geometric time resolution of ~ 1 as.

The experiment was performed at BL3 of SPring-8 Angstrom Compact free-electron LAsER (SACLA) [7,23]. Two 10-keV XFEL subpulses with a variable delay were generated through an x-ray autocorrelator employing Si(220) crystals with a reflection bandwidth of 0.56 eV [20,24]. The subpulses were focused by a Kirkpatrick-Baez mirror system [25,26] and spatially overlapped onto a (001)-faced single crystal diamond with a thickness of 500 μm . The beam size was 1 μm (horizontal) $\times$ 3 μm (vertical) in full width at half maximum (FWHM). A small crossing angle of

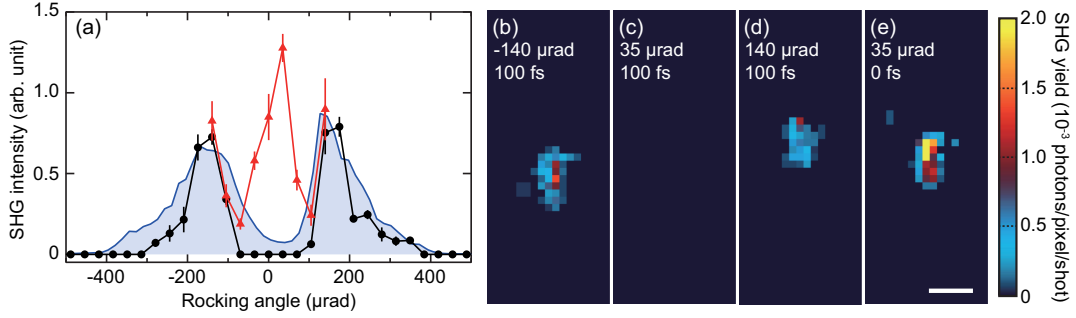


FIG. 2. (a) SHG rocking curves at $\Delta t = 0$ fs (triangle) and 100 fs (circle). 10 000 (20 000) shots were accumulated for each angle at 0 fs (100 fs). Error bar denotes the fitting error. Solid line filled to zero shows a rocking curve of the diamond 220 reflection at 10 keV. Spatial distributions of detected second-harmonic photons $\Delta t = 100$ fs (b)–(d) and 0 fs (e). The reflection plane is along vertical. Rocking angles are displayed in each panel. Scale bar denotes 300 μm .

~ 300 μrad was introduced between the two pulses in the horizontal direction, as shown in Fig. 1. This was confirmed by the rocking curve of the Laue-case diamond 220 reflection in the horizontal plane at 10 keV [solid line filled to zero in Fig. 2(a)], where two peaks were observed, and the angular separation between them represents the crossing angle. The average pulse energies were ~ 3 $\mu\text{J} + 3$ μJ (total 6 μJ), corresponding to a peak intensity of $> 10^{16}$ Wcm^{-2} and a deposited energy of ~ 0.05 eV/atom, which is well below the diamond damage threshold of ~ 1 eV/atom [27]. The phase-matching condition of x-ray SHG was satisfied through the diamond 220 reflection, and the generated second-harmonic photons were counted by a multiport charge-coupled device (MPCCD) [28] located 1 m away from the diamond. A key advantage of SHG over multiphoton excitation processes, such as TPA, is that the signal is concentrated in a much smaller solid angle, enabling efficient detection with a small detector [see Figs. 2(b)–2(e)]. The detection efficiency was estimated to be $\sim 11\%$, resulting in an efficiency ~ 100 times higher than that reported in [20].

SHG IAC measurements were conducted under two conditions: (1) without and (2) with an additional Si(311) double channel-cut crystal monochromator (DCCM) [29]. The Si(311) DCCM reduced the bandwidth of the subpulses from 0.56 to 0.27 eV, which is narrower than typical spectral spike widths of 0.3–0.4 eV FWHM at SACLA [17,18,24]. In this case, both the amplitude and phase of the spectrum could be dominated by the response function of the four-bounce Si 311 reflections [20], stabilizing the temporal profile of output pulses. Here, the vertical focus size was reduced to 1 μm to compensate for the threefold decrease in pulse energy (1 $\mu\text{J} + 1$ μJ) caused by the reduced bandwidth. See Appendix A for additional information on the experimental setup and procedure.

Figure 2(a) shows rocking curves of the SHG process measured at time delays $\Delta t = 0$ fs and 100 fs. Vertical axis represents the SHG intensity, which is a coefficient a of a quadratic function $y = ax^2$ fitted to the SHG yield (y) as a function of the summed pulse energy (x) (see Appendix B for details). Three peaks are observed at $\Delta t = 0$ fs.

Second-harmonic photons generated at the three peak angles spatially overlapped to each other [Figs. 2(b)–2(e)], due to the small crossing angle. The central peak disappeared at $\Delta t = 100$ fs, whereas the other two remained, indicating that the two side peaks are the delay-independent SHG signals generated through Eqs. (1b) and (1c), and the central one is the autocorrelation signal through Eq. (1a). The width of each peak was 100–150 μrad FWHM. Remarkably, no signal was observed at $\Delta t = 100$ fs during 100 000-shot acquisition at the central peak angle, confirming the validity of the background-free concept proposed in this Letter.

Figure 3(a) presents an IAC trace measured at the central peak angle in Fig. 2 without the Si(311) DCCM. Owing to the zero background and the high SHG detection efficiency, much smaller uncertainties were achieved with less acquisition—70 000 shots at each delay—compared to > 400 000 shots in the TPA-IAC measurement [20]. Two features appeared in the trace: a narrow central peak and a broader wing. These features likely originate from shot-to-shot variations of temporal profiles after the x-ray autocorrelator because the IAC measurement yields an averaged temporal profile. Indeed, independently measured spectra of the subpulses showed significant variations affecting the original ones with a spike width ranging from 0.15 to 0.35 eV FWHM, which is narrower than the Si(220) reflection bandwidth of 0.56 eV. Consequently, each spectrum contained one to three randomly distributed spikes.

Numerical simulations indicate that multispike events produce multiple temporal spikes of which autocorrelation functions include side peaks (see Appendix C for details). Since the peak locations reflect the temporal separations between spikes, they become blurred in the averaging process, forming a broad wing in the final autocorrelation trace. The width of the main peak reflects the duration of each temporal spike, which can be dominated by the impulse response of the x-ray autocorrelator. In fact, the central peak agrees well with the autocorrelation of the impulse response (thick dashed line). In contrast, single-spike events contain only one temporal spike with

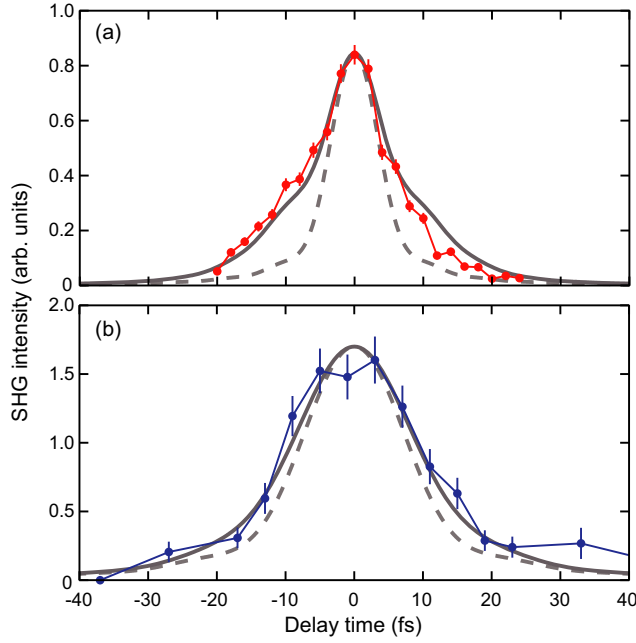


FIG. 3. Measured intensity autocorrelation traces (circles) without (a) and with (b) the Si(311) monochromator. Error bar denotes the fitting error. Number of accumulations were 70 000 and 60 000 shots at each point for plots (a) and (b), respectively. Thick dashed lines are the autocorrelation functions of the calculated impulse responses and the thick solid lines the averaged ones of simulated temporal profiles for each configuration.

a duration longer than the impulse response, resulting in a relatively broad single-peak autocorrelation. The measured trace can thus be interpreted as a superposition of these different autocorrelation contributions. Consistently, the averaged autocorrelation obtained from the simulated temporal profiles (thick solid line) shows good agreement with the measured trace.

On the other hand, most output pulses contained only a single spectral spike when the Si(311) DCCM was used. The measured IAC trace [Fig. 3(b)] shows a single peak and reasonably agrees with the autocorrelation of the impulse response (thick dashed line). This result strongly suggests that the temporal profile of the XFEL pulses was modified by the Si 311 reflections, forming a well-defined profile. Note that the measured autocorrelation width of 23 fs FWHM was slightly wider than the calculated width of 18 fs FWHM for the impulse response. As mentioned above, the spectral spike width (0.15–0.35 eV FWHM) was comparable to or narrower than the Si(311) reflection bandwidth of 0.27 eV. Therefore, the spectrum of the output pulses more closely resembled the original spectrum within the reflection bandwidth rather than the response function of the monochromators, resulting in a pulse duration longer than that of the impulse response. Similarly to the former case, the numerical simulation (thick solid line) well reproduces the measured trace.

Background-free IAC measurements were successfully achieved for narrowband femtosecond x-ray pulses, using SHG operating in a crossing geometry. Here, we discuss the applicability of this method to broadband XFEL pulses, particularly attosecond pulses with a bandwidth greater than 10 eV. Although the narrow phase-matching condition of the SHG process facilitates the exclusion of background, it may limit the acceptable bandwidth of the incident wave. First, we show the angular and bandwidth acceptances for the SHG process using analytical expressions for a Gaussian beam described in [22]. The SHG efficiency η_{SH} decreases with a phase mismatch, $\Delta k = |\mathbf{k}_{2\omega} - \mathbf{G} - 2\mathbf{k}_{\omega}|$, caused by angular and frequency detuning. Neglecting absorption in the medium, the efficiency can be expressed by

$$\eta_{\text{SH}} \propto I_{\omega} L_{s-t} L \exp \left[- \left(\frac{\Delta k L_{s-t}}{2} \right)^2 \right], \quad (2)$$

where I_{ω} is the intensity of the fundamental wave that is inversely proportional to the lateral beam size and pulse duration, L_{s-t} is the effective interaction length between the fundamental and second-harmonic waves, and L is the crystal thickness. Equation (2) indicates that the Δk acceptance is inversely proportional to L_{s-t} . Note that L_{s-t} is typically much smaller than L under the walk-off condition where the propagation direction of the second-harmonic wave is noncollinear to that of the fundamental wave, causing spatial separation during propagation. Here, since $\mathbf{k}_{2\omega}$ is also not parallel to $\mathbf{k}_{\omega}$, the effective group velocity in diamond of the second-harmonic wave along $\mathbf{k}_{\omega}$ can be expressed as $v_{g,2\omega} \cos \rho$, where $v_{g,2\omega}$ is the characteristic group velocity of the second-harmonic wave, and ρ , the walk-off angle, can be approximated as $2\theta_B$ with the geometric Bragg angle θ_B for the second-harmonic wave. By considering these walk-off effects, L_{s-t} can be defined as

$$L_{s-t} = \left(\frac{\tan^2 \rho}{w^2} + \frac{\text{GVM}^2}{\tau_{\text{TL}}^2} \right)^{-1/2}, \quad (3)$$

where w is the beam size in the reflection plane, τ_{TL} is the transform-limited pulse duration, which is inversely proportional to the bandwidth, and $\text{GVM} = (v_{g,2\omega} \cos \rho)^{-1} - v_{g,\omega}^{-1}$ is the group velocity mismatch, where $v_{g,\omega}$ denotes the group velocity of the fundamental wave. As we accept Δk at which η_{SH} becomes half of its maximum, the FWHM angle of the acceptance, which is the intrinsic rocking-curve width, can be expressed as

$$\Delta \theta_{\text{FWHM}} = \frac{2\sqrt{\ln 2}}{2\pi k_{\omega} L_{s-t} \tan(2\theta_B)}. \quad (4)$$

Similarly, the bandwidth acceptance can be approximated by

$$\Delta E_{\text{FWHM}} = \frac{\sqrt{\ln 2} \cos(2\theta_B)}{L_{s-t} \sin^2 \theta_B} \hbar c, \quad (5)$$

where $\hbar$ is the reduced Planck's constant.

Using Eqs. (3)–(5), we estimated $\Delta\theta_{\text{FWHM}}$ and ΔE_{FWHM} to be 57 μrad and 2.3 eV, respectively, under the current experimental conditions with the beam size of 1.0 μm FWHM ($w = 0.60 \mu\text{m}$) and the bandwidth $\Delta E_{\omega} = 0.56 \text{ eV}$ FWHM ($\tau_{\text{TL}} = 2.0 \text{ fs}$). Here, the interaction length L_{s-t} was mainly limited by the small beam size, making the bandwidth acceptance wider than ΔE_{ω} . Remarkably, even if the beam size is infinite and, consequently, $L_{s-t} = \tau_{\text{TL}}/\text{GVM}$ where the bandwidth acceptance is the narrowest, it is still slightly wider than ΔE_{ω} . This fact suggests that the SHG process has almost no limitation in ΔE_{ω} . However, the rocking-curve width $\Delta\theta_{\text{FWHM}}$ also becomes wider for broadband waves, necessitating a larger crossing angle for background exclusion while degrading the geometric time resolution.

By assuming $\Delta E_{\omega} = 50 \text{ eV}$ FWHM, corresponding to $\tau_{\text{TL}} = 22 \text{ as}$, and the same beam size as the present experiment, $\Delta\theta_{\text{FWHM}}$ is estimated to be 1.3 mrad, which requires a crossing angle $\alpha > 3 \text{ mrad}$ to prevent SHG signals from overlapping, degrading the geometric time resolution to $> 10 \text{ as}$. Such a large crossing angle is readily available by means of a 100-nm focusing mirror at SACLA [30]. At the small beam size of 100 nm FWHM, a 1-as geometric time resolution can also be achieved even at $\alpha = 3 \text{ mrad}$. The reduction of the SHG efficiency due to the short interaction length can be compensated by the enhanced intensity (see Appendix D). Therefore, the background-free SHG IAC measurement should be feasible for attosecond XFEL pulses if an autocorrelator with broadband optics, such as total reflection [31] or multilayer-coated mirrors [32], is employed instead of narrowband crystal optics.

In summary, this Letter demonstrated that background-free IAC measurements, operating in a crossing geometry, can be achieved using a wave-mixing process while realizing a geometric time resolution down to 1 as. It was verified that x-ray pulse shape was modified, and possibly stabilized, by a narrowband monochromator. The excellent radiation hardness of diamond enabled nondestructive measurements, which is particularly advantageous for high-repetition-rate XFEL sources, as fast sample scanning is not required. A broadband autocorrelator combined with a nanofocusing device would allow characterization of broadband attosecond x-ray pulses without any background. Moreover, IAC measurements using wave-mixing processes, which are sensitive to the phase of incident waves, can be extended to frequency-resolved optical gating [33], paving the way toward model-free full characterization of ultrashort x-ray pulses with durations down to a few attoseconds.

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Data availability—The data that support the findings of this article are not publicly available. The data are available from the authors upon reasonable request.

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End Matter

Appendix A: Experimental details—We provide additional details of the experimental setup and procedure. XFEL pulses at 10 keV were generated via the reflection self-seeding scheme [34] at a repetition rate of 30 Hz. Second-harmonic radiation from the source was suppressed using a Si(111) double-crystal monochromator, followed by the x-ray autocorrelator equipped with Si(220) crystals and the Si(311) DCCM. Shot-to-shot spectra downstream of these monochromators were recorded destructively with a high-resolution dispersive spectrometer consisting of an elliptical focusing mirror, a Si(660) flat-crystal analyzer, and an MPCCD detector [29]. The spectral resolution was 43 meV per pixel. In addition to the spectral measurements, the spectrometer was used to match the photon energies of the pulse replicas.

Prior to the IAC measurements, we measured a rocking curve of the diamond 220 reflection in the Laue geometry at 10 keV to determine the crossing angle between the subpulses. A photodiode positioned at a scattering angle of 59° detected the Bragg-reflected photons from both the subpulses. A 0.2-mm-thick Si attenuator reduced the incident x-ray flux to $\sim 20\%$ to avoid saturating the diode. The crossing angle was evaluated to be $\sim 300 \mu\text{rad}$, as shown in Fig. 2 of the main text.

Second-harmonic photons were detected using an MPCCD with a 300- μm -thick Si sensor (Phase-IIIL) located 1 m away from the diamond crystal. The quantum efficiencies at 10 and 20 keV were 89% and 29%, respectively. A 0.8-mm-thick aluminum filter located in front of the detector suppressed elastically scattered 10-keV photons to 0.5%, while transmitting $\sim 50\%$ of the 20-keV photons. Including the 1-m air path, the resulting effective detection efficiencies were estimated to be 0.25% and 11% for 10- and 20-keV photons, respectively.

A photodiffusion-free transparent scintillator detector was employed to optimize the spatial overlap of the pulse replicas at the sample plane. The detector comprises a 5- μm -thick Ce-doped $\text{Lu}_3\text{Al}_5\text{O}_{12}$ scintillator bonded to an undoped $\text{Lu}_3\text{Al}_5\text{O}_{12}$ substrate [35], coupled to a $20\times$ objective lens. The effective spatial resolution was better than 1 μm , with a pixel size of 0.32 μm . Spatial overlap between the two subpulses was finely tuned by adjusting the beam angle of one of the subpulses (called the delay-variable branch; see Ref. [20] for details) at a tens of nanoradian precision.

IAC traces were obtained by repeating time-delay scans. At each delay point, 5000 shots were acquired, and each full scan required approximately 30 min. Spatial overlap was verified every three or four scans.

Appendix B: Calculation of SHG intensity—Here, we describe the details for calculating the SHG intensity. A small region of interest of 1.5 mm (30 pixels) in the

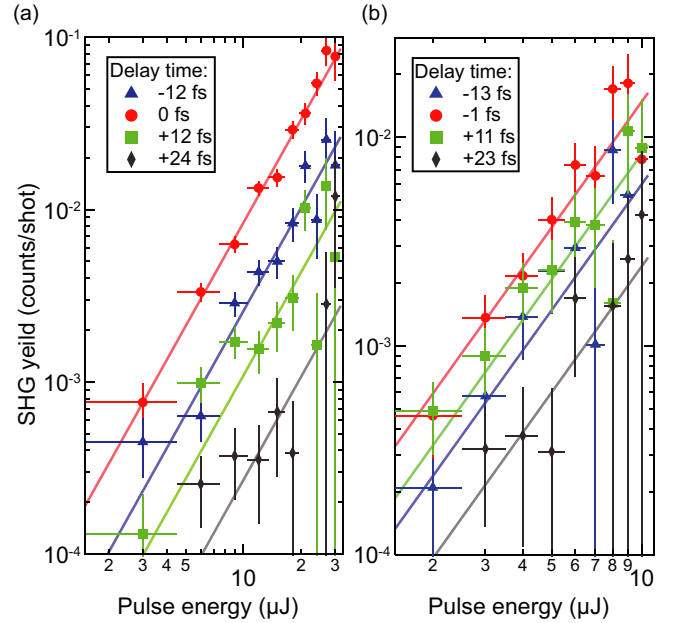


FIG. 4. SHG-yield dependence on pulse energy at several delay times measured without (a) and with (b) the Si(311) DCCM. The delay time for each plot is displayed in the figure. Vertical and horizontal error bars are statistical errors and bin widths, respectively. Solid lines represent the fitted quadratic functions.

azimuthal direction $\times 5 \text{ mm}$ (100 pixels) in the radial direction around the expected location of SHG signals was defined on the MPCCD detector. The MPCCD images were analyzed using a droplet algorithm [36], which reconstructs individual x-ray photon events by collecting the dispersed charge associated with each photon. The photon energy of each droplet was determined from the summed detector counts, after which photons with energies ranging from 17 to 23 keV were identified as second-harmonic signals.

All shots were sorted into bins according to the summed incident pulse energy, and the SHG yield was calculated for each bin. Representative examples of the SHG-yield dependence on pulse energy are shown in Fig. 4. As described in the main text, each SHG yield was fitted with a quadratic function $y = ax^2$, where y is the SHG yield and x is the incident pulse energy. The fitted coefficient a was taken as the SHG intensity.

In this analysis, we assumed that the spatial properties of the two subpulses remained unchanged throughout the measurements, and that only the pulse energy, temporal and spectral properties varied shot by shot.

Appendix C: Numerical simulation of temporal profiles—We numerically reconstructed temporal profiles and spectra of the original XFEL pulses following the procedure described in [37]. First, the electric field in

TABLE I. Parameter settings for simulation.

	A_i	t_i	$\Delta\omega_i/\omega_0$	ϕ_i	σ_{spike}	n
Distribution	Uniform	Normal	Uniform	Uniform	Constant	Constant
Range	0.75–1.0	5.73 fs ^a	$\pm 6.5 \times 10^{-5}$	$\pm\pi$	0.34 fs ^a	200

^aStandard deviation.

the time domain, $E_{\text{in}}(t)$, was generated as

$$E_{\text{in}}(t) = \sum_{i=1}^n A_i \exp\left[-\frac{(t-t_i)^2}{4\sigma_{\text{spike}}^2}\right] \times \exp\{i[(\omega_0 + \Delta\omega_i)(t-t_i) + \phi_i]\}, \quad (\text{C1})$$

where A_i , t_i , $\Delta\omega_i$, and ϕ_i are the amplitude, occurrence time, angular-frequency deviation from ω_0 , and phase at t_i of the i th temporal spike. The parameter σ_{spike} corresponds to the spike duration, and n is the initial number of assigned spikes. Note that the actual number of spikes can be smaller than n , because spikes with nearly identical occurrence times merge into a single structure.

The corresponding frequency-domain field was obtained via Fourier transformation,

$$E_{\text{in}}(\omega) = \int E_{\text{in}}(t) e^{-i\omega t} dt \quad (\text{C2})$$

To reproduce the experimentally observed spectra, we chose parameters as summarized in Table I. The wave field was then modified by multiplying the amplitude reflectivities in the frequency domain of the Si 111 [$R_{111}(\omega)$], Si 220 [$R_{220}(\omega)$], and Si 311 reflections [$R_{311}(\omega)$] calculated through dynamical theory of x-ray diffraction [38],

$$E_{\text{out}}(\omega) = E_{\text{in}}(\omega) R_{111}^2(\omega) R_{220}^4(\omega) \quad \text{or} \quad E_{\text{in}}(\omega) R_{111}^2(\omega) R_{220}^4(\omega) R_{311}^4(\omega), \quad (\text{C3})$$

depending on the monochromator configuration. Inverse Fourier transformation of $E_{\text{out}}(\omega)$ yielded the temporal wave fields $E_{\text{out}}(t)$ of the pulse replicas. The impulse

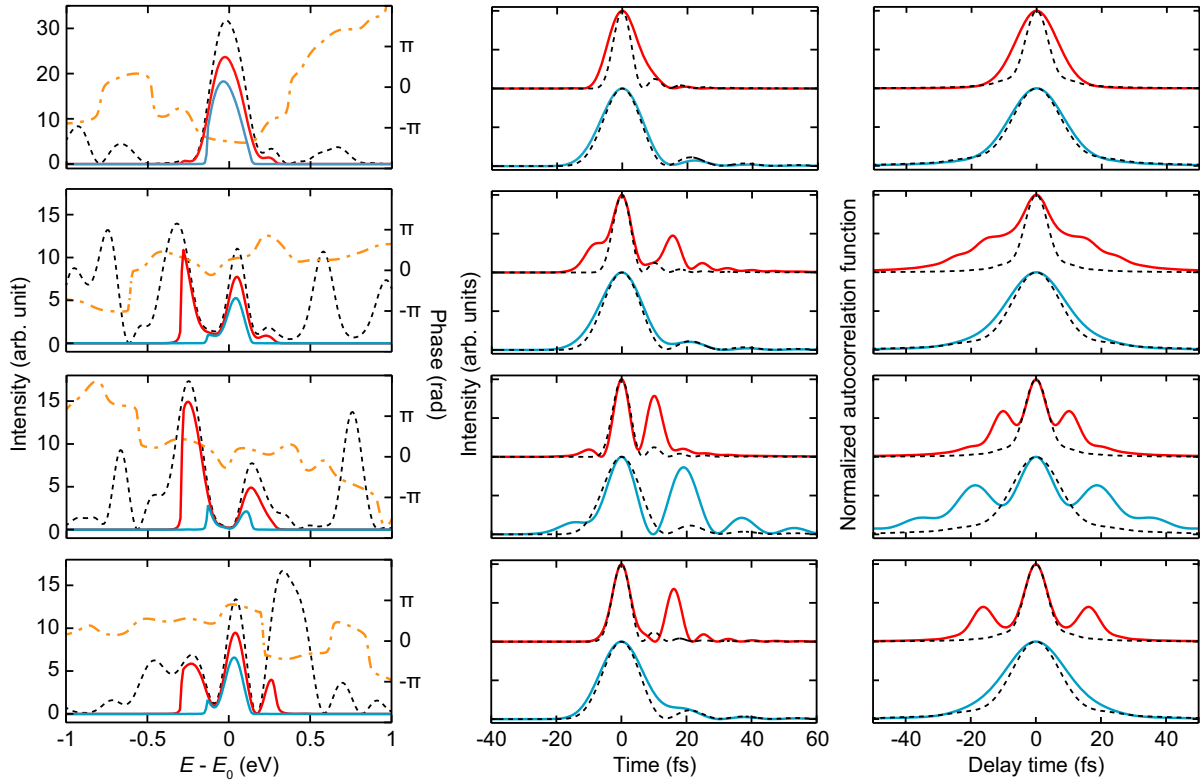


FIG. 5. Examples of simulated spectra, temporal profiles, and the corresponding autocorrelation functions. Left column presents simulated spectra of the original (dashed line), downstream of the x-ray autocorrelator (red line), and the Si(311) DCCM (blue line). The dash-dotted line is the phase of the original spectra (right axis). Middle and right columns show temporal profiles calculated from each spectrum and their autocorrelation functions, respectively. Dashed lines represent impulse responses and their autocorrelation functions.

responses without and with the Si(311) DCCM correspond to the Fourier transforms of $R_{111}^2(\omega)R_{220}^4(\omega)$ and $R_{111}^2(\omega)R_{220}^4(\omega)R_{311}^4(\omega)$, respectively.

We generated 1000 wave-field patterns, representative examples of which are shown in Fig. 5. In the first-row example, only a single spectral spike lies within the Si(220) reflection bandwidth, and the corresponding temporal profile exhibits a single peak with a duration longer than the impulse response. In other cases, two or three spikes are present, producing multiple temporal spikes in the absence of the Si(311) DCCM. In the third- and fourth-row examples, the duration of each temporal spike is consistent with the impulse response, resulting in a central autocorrelation peak similar to that of the impulse response along with side peaks. In contrast, except for the third example, only one spectral spike remains within the Si(311) bandwidth, producing temporal profiles and autocorrelation functions that are broader than the corresponding impulse response.

For each configuration, we selected the 200 patterns with the largest summed amplitudes, $\int |E_{\text{out}}(\omega)|^2 d\omega$, and averaged their normalized autocorrelation functions. The averaged autocorrelations show good agreement with the measured IAC traces, as presented in Fig. 3 of the main text.

Appendix D: Estimation of SHG efficiency for nanofocused broadband beams—Here, we estimate the SHG efficiency for a 100-nm broadband ($\Delta E_\omega = 50$ eV FWHM) beam, whose interaction length is much shorter than that of the 1- μm narrowband beam used in the experiment, as described in the main text. According to Eq. (2), the efficiency scales as L_{s-t} , giving a reduction by a factor of 25. In addition, the crystal thickness L must be sufficiently thinner than the focal depth, which is ~ 45 μm for a 100-nm beam at 10 keV. If we assume $L = 20$ μm , the efficiency decreases by another factor of 25, resulting in an overall reduction by a factor of 625.

At the same time, however, the small focal spot and ultrashort pulse duration substantially increase the intensity I_ω . By assuming a beam size of 100 nm $\times$ 3 μm FWHM, a pulse duration of 50 as FWHM, and the same x-ray flux as the present experiment, I_ω increases by a factor of $\sim 2 \times 10^3$ with a deposited energy of ~ 0.5 eV/atom, which is still below the diamond damage threshold. Consequently, the SHG efficiency as well as yield for a 100-nm broadband beam could be comparable to—or even higher than—that achieved in the present experiment while avoiding damage on the diamond crystal.